

# Alexander DeLise

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## EDUCATION

### Florida State University

Tallahassee, FL

*B.S. Applied and Computational Mathematics, B.S. Computational Science, GPA - 4.0*

*Aug 2023 - May 2027*

- **Honors/Awards:** Presidential Scholar, Honors Scholar, Vires Scholar, Seminole Scholar (x2), Bright Futures Scholar, GGK Torch Night Award Recipient, President's List (x3)
- **Relevant Coursework:** Computational Probabilistic Modeling, Discrete/Continuous Algorithms, Programming for Scientific Applications, Math Modeling, Statistics for Engineers

## EXPERIENCE

### FAMU-FSU College of Engineering

Sep 2024 - Present

*Researcher, Operations Research*

*Tallahassee, FL*

- Project title: Data-Driven Patient Allocation Optimization with Epidemic and Vaccine Modeling; first author on paper accepted to the *2025 North American Conference on Industrial Engineering and Operations Management*, also presented at the *2025 Florida Undergraduate Research Conference*
- Developed a mixed-integer nonlinear operations research model integrating the Susceptible, Infected, Recovered, Vaccinated framework with patient allocation and transfer dynamics
- Simulated patient allocation scenarios using Florida Department of Health data during the COVID-19 pandemic, highlighting reductions in unmet demand and patient transfers across decision periods

### University of Tennessee, Knoxville

May 2024 - Present

*Researcher, Quantum Algorithms*

*Knoxville, TN*

- Project title: Learning Feasible States with QAOA and ma-QAOA for Constrained Optimization, presented at *2025 Joint Mathematics Meetings Conference* and *2024 Summer Research Scholars Symposium* at UTK
- Developed a novel approach to implementing constrained optimization problems on quantum hardware through ancillae state preparation with accuracy of over 99% on low-qubit quadratic optimization problems
- Implemented methods using PennyLane, a quantum machine learning Python API

### FAMU-FSU College of Engineering

Sep 2023 - Apr 2024

*Researcher, Operations Research*

*Tallahassee, FL*

- Project title: Cost-Effective Location-Allocation for COVID-19 Patient Assignment in Florida: A Data-Driven Approach, presented at *2024 Undergraduate Research Symposium* at FSU
- Developed a linear operations research model to simulate location-allocation of Florida COVID-19 patients subject to budget and travel constraints using Florida Department of Health and geospatial data
- Determined optimal location to establish new healthcare facilities in Florida under various scenarios
- Harnessed IBM ILOG CPLEX optimization software and R for data visualization

## PROJECTS

### Support Vector Regression (SVR) for Stock Price Prediction | *Python, TensorFlow, Scikit-learn* April 2024

- Engineered SVR models using **Scikit-learn** with linear, radial basis function, and polynomial kernels to optimize 14-day stock price predictions
- Integrated TensorFlow to benchmark SVR against DNNs, with SVR achieving over 50% greater accuracy
- Authored technical report detailing kernel optimization, model performance metrics, and strategies for future enhancement in financial forecasting models

### Adaptive Randomized Range Finder | *Python, PyTorch, NumPy, Matplotlib*

February 2024

- Integrated the Adaptive Range Finder Algorithm in Python with PyTorch for high dimensional matrix rank and range approximation
- Applied random Gaussian vectors and orthonormal projections to optimize matrix decomposition
- Implemented continuous delivery using TravisCI to build the plugin upon new a release
- Achieved average approximation error of less than  $10^{-4}$ , with performance visualized using Matplotlib

## TECHNICAL SKILLS

**Languages:** Python, R, MATLAB, C++

**Tools:** Git, GitHub, VS Code, IBM ILOG CPLEX, GAMS, Gurobi, Microsoft Office Suite

**Frameworks and Libraries:** pandas, NumPy, Matplotlib, SciPy, scikit-learn, Gurobi API, PennyLane API